

Exploratory Data Analysis on the Penguin Size Dataset

Report

Introduction

This dataset tracks the size differences among 3 penguin species across different islands. It contains 344 data points and 7 columns.

Data cleaning

Determining the type of values that are researched helps clean and determine the type of data required. A scatter plot was used to compare varying information. As the data between islands did not produce a major difference, it was removed from the data.

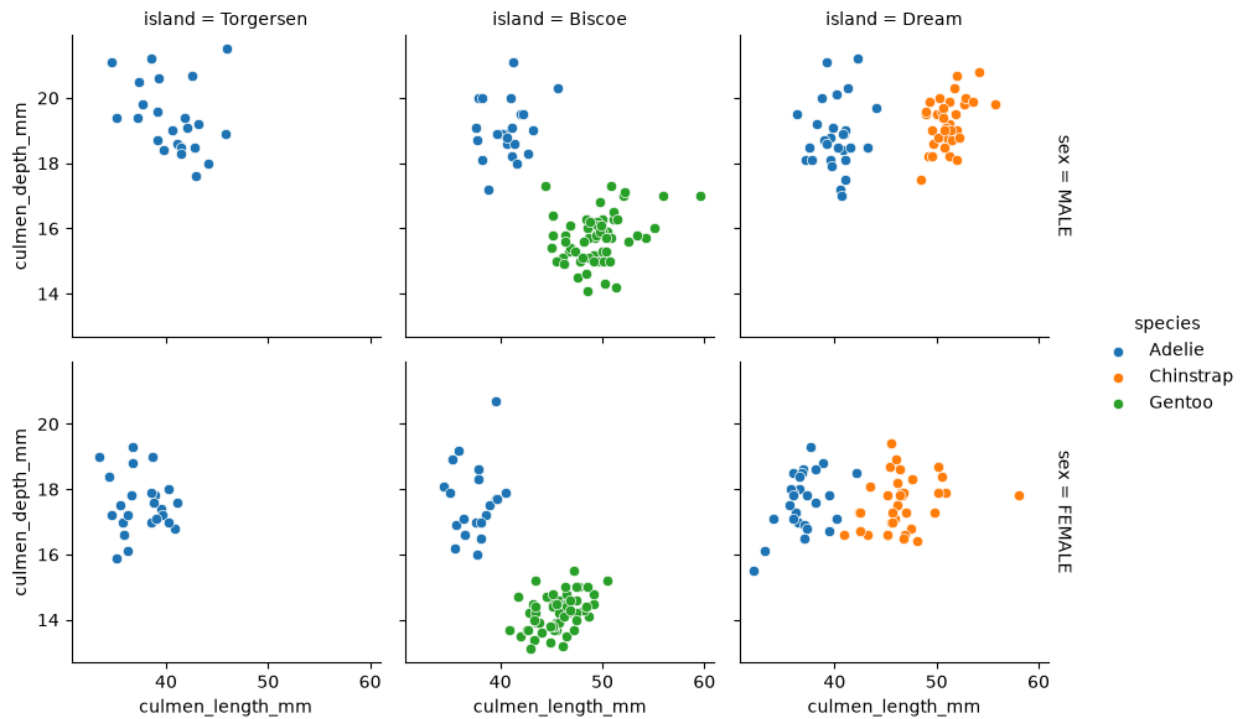
Missing data

There is some missing information in the data. Depending on the specific value that is being researched, the missing information could be ignored. However, some of the penguins are missing almost all information; therefore, the subject is not included in the data. Penguins without data are not shown on the graph, and penguins with unknown genders are removed.

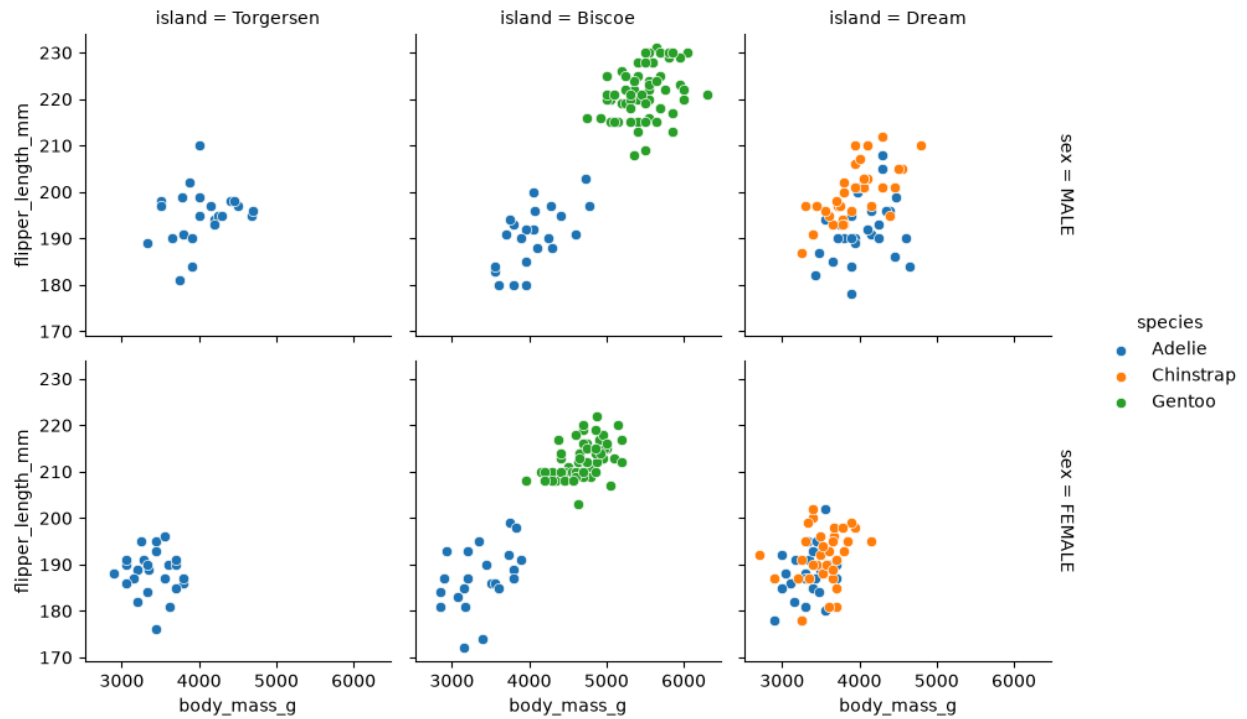
Data stories and visualisations

The biggest differentiation between the penguins is the size difference between males and females. Males are always bigger than females in all areas, including flipper length, beak depth

and size, and body mass. There is a positive correlation between beak sizes, body mass, and flipper lengths. Island location does not affect the Adelle penguin population.



For beak sizes, the length shows both positive, negative, and neutral correlations. In male Adelles on Torgersen island shows a negative correlation, where a longer beak decreases beak depth, while female Adelles on Dream Island show a positive correlation. Other than these observations, the length does not correlate with beak depth. Chinstraps and Gentoos also demonstrate a neutral correlation regarding beak length and depth.



Flipper size and body mass have a positive correlation no matter the breed. The heavier body mass has longer flippers. These variables also demonstrate that males are bigger than females. Gentoos are overall bigger than Adelles and Chinstrap. Cihnstraps are about the same size as Adelles.

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